MA-300/MA-350/MA-360/MAW300/MA-M00

Module Handbook Undergraduate Project in Mathematics

SEPTEMBER 1, 2025

Chapter 1: General Information on the Project

1.1. Overview

For this module you are tasked with carrying out a project. That project has a number of goals that you need to achieve, a number of deadlines, and constraints in the support and resources available to you. This is designed to exactly mirror the sort of situation most of you will encounter regularly when you have a professional job – a set of inter-related tasks, with deadlines and constraints.

The primary goal that you need to achieve in this project is to produce a written report of 7000-8000 words in length on a prescribed area of mathematics that you have researched and taught yourself.

There are also six secondary goals:

1. To complete some initial reading/tasks provided by your supervisor and then present and discuss your work and plans with your supervisor.
2. To write a preliminary "Project Plan" summarizing your plan to manage the project.
3. To give an interim presentation on your achievements part way through the project.
4. To submit an initial draft of a section of your report so that you may receive some useful feedback on your writing.
5. To reflect upon what you have learned, how you worked, and how you adapted your original "Project Plan".
6. To give a final presentation on your final written report once this is completed.

The exact deadlines and requirements for these six goals will be given at the start of the year; it is your responsibility to make sure you know these deadlines, and to plan your work to achieve them on time. The information you need will be on Canvas and/or will have been given in lectures. It is your responsibility to take in this information for when you need it.

1.2. Skills required

Clearly this is quite a different module compared with most of those you have taken so far. You will need some different skills, some of which you will already have, and some of which you may need to acquire. The skills you acquire, and demonstrate through this module, will help you in whatever you choose to do next. These are *transferable* skills that can be applied in many different situations.

Throughout the year we will provide a number of lectures that will equip you with the skills you need. There will be lectures on time management, on choices, on planning, on presentations, on referencing, among other topics. The whole project is about developing skills you will need in your future career, and some of the lectures will be more explicitly about this – focussing on how to choose which jobs to apply for, how to write a good CV, how to perform well in interview etc. Some are about the immediate goals of the project and some are longer term. This, again, is giving you an opportunity to develop an essential skill: that of taking in information when you are given it, and storing it until you need it.

In addition to the lectures there will be written material, such as these notes, which will also help you develop the skills you need.

You will also have meetings with your project supervisor, but those are to give you the mathematical guidance specific to your topic. If you have a question about a theorem that you are thinking about including, then you should ask your supervisor. But if you have a question about the length of the written report, or the format of the presentations, or any other general questions that apply to all of the students on the module, you should ask in one of the lectures, so that all students may benefit from the answer.

1.3. Independence

One of the most important skills you will develop through this module is the ability to work independently. You alone are responsible for achieving the goals, so you must motivate yourself to do the work required for this, to do enough work, and at the appropriate time. Your supervisor is there to provide mathematical support, and answer mathematical questions, but you are given the independence to work at your chosen speed, and in your chosen way, so long as you achieve the goals by their deadlines.

This means you must be **proactive**. It is not enough to sit and wait to be told what to do, and when to do it. We will tell you the deadlines at the start of the year; it is your responsibility to take in this information and act accordingly. We may not remind you that a deadline is looming next week – you need to remember when the deadlines are, or know how to find them online.

And you must act in a timely manner. A good written report cannot be put together in just a few days before the deadline, so you must plan ahead. And you should be prepared for unexpected disasters: if your computer crashes the day before you go to print your work, you must take steps to ensure that this does not affect your project, just as if you want to buy a lottery ticket on Saturday night, it's no use turning up on Sunday morning saying that the bus was late. Whatever technology you use, you must be prepared for problems – computers do die, files can get corrupted, laptops can get stolen. But all of these problems can be mitigated – save your work in multiple places, email it to yourself, etc. Always make sure you have back-ups stored elsewhere, and have a plan B (and maybe even a plan C and a plan D) for when you problems occur.

Of course, there are some extenuating circumstances that are beyond your control but which may prevent you from meeting the deadlines. The University policy on extenuating circumstances can be found on its website and gives details of what sort of circumstances may justify an extension. If you feel you are in such circumstances, then let us know and we will discuss what adjustments we can make.

1.4. The role of your supervisor

The project is your work and your responsibility, and you have a lot of independence to carry this work out in your own way. Nevertheless, you have been assigned a supervisor who is there to support you along the way. Their role is primarily as a mathematical expert, to assist you with any questions you have about your topic.

If you cannot understand the proof of a certain theorem, despite all your efforts, then your supervisor can help. If you cannot find a good source for a certain topic, then your supervisor can help. If you have questions about how to submit your work, or what software to use for your presentation, then you should ask the Project Coordinator. If you cannot remember the deadline for the next goal in your project then .... you need to take responsibility and find this information out for yourself – do not expect your supervisor to help you in this.

When you are writing up your report, you may wish to ask for feedback and advice on your work. Your supervisor will always be able to identify things in your project that could be improved; there is no such thing as a perfect project. And when your supervisor points out something that can be improved, it is up to you to decide how to change it, and what to replace it with – your supervisor cannot tell you that a) because then they would be writing the project not you and b) because there is no perfect answer for what should be written. In many mathematics assignments there is one perfect answer, but that simply does not apply to this sort of written project, nor to an oral presentation.

Clearly if you see your supervisor every week for an hour, and another student only sees their supervisor once during the whole year, then this would be unfair. Similarly if you submit ten different drafts to your supervisor, getting comments on each one, then that would be unfair compared with a student who only submitted one. So we have to place some controls on the amount of support you get from your supervisor in order to ensure fairness as much as possible for different students.

The supervisor will generally mark at least one of your presentations, and will be one of the markers for your final written report. Their marking of your report will take into account the amount of support you've received.

You can expect your supervisor to

- advise you about books and sources to use during your project,
- offer advice about the content and direction of your report,
- offer help with understanding mathematical issues,
- give feedback on how you can improve your work.

You should not expect your supervisor to

- write your project for you,
- supply you with all the sources you need to research your topic,
- tell you exactly what you need to write to achieve a certain grade,
- warn you of impending deadlines.

Chapter 2: Primary Goal: The Written Report

2.1. Introduction

By the end of this project you will have written a report on an area of mathematics that you have taught yourself, beyond what you have covered in your taught modules. This report should be produced to a professional standard, typed in LaTeX. The subject of your report will be an area of mathematics that you will, when you have completed the report, be something of an expert on. It will have become *your* area of mathematics.

What do we expect the content of this report to be? It should be a summary of one area of mathematics, written so that someone who does not know the area can read your report and learn the subject.

It is not intended that you carry out research in mathematics and create/discover new mathematics – that is PhD-level work and not realistically achievable for an undergraduate student. Similarly, it is not intended that your project simply rehashes lecture notes for courses you have already taken. Instead, you are to find material in a variety of sources, compile it together into your own description of the topic.

It must be *your own work*. Even though everything in your report will be based on other sources, you must rewrite the material in your own words, and make it your own. If you do not rewrite the material, then you risk committing academic misconduct or plagiarism, for which the punishments can be severe, including the possibility of being withdrawn from the University. There is more information about how to avoid plagiarism below.

2.2. Getting started

Your first task has already been to select two areas of mathematics that you are interested in. We have used this to allocate you a supervisor with the aim of ensuring that everyone can complete a project in one of the areas that they have selected.

Now a supervisor has been allocated, you will have a first meeting at which your supervisor will discuss with you the project that you will work on in your chosen area. You will agree on a title and outline for the project, and they will provide you with some initial reading that you must complete.

The aim of this initial reading is to introduce you to the topic that you will be working on for the next year. This will help you to get an idea of the work that you will need to complete over the forthcoming year. It is essential that you do this to establish a background in the subject which will be necessary to begin planning what you are going to do over the year.

You should begin by reading the sources you have been given, and from this you should be able to form an overall impression of the topic – its main themes, how it relates to other areas of mathematics and science, what are the key results, etc. And you should start to seek out other sources on the subject.

You can then begin to work out the scope of your project, what you will cover and what you will omit, and what the sources are that you are likely to use. You can start to discern the main themes that you want to emphasize in your report. But be aware that all this will change and evolve as you learn the topic better.

Once you have this overview, you will be expected to meet with your supervisor to present to them the work you have done, and what you are planning to cover in your project. They will then help you to turn this into an effective plan for the completion of your project. You should have an idea of how many chapters there will be, which will be longer and which will be shorter, which will take more time to research and write and which will take less time. At this point you should plan your work schedule for the remainder of the project. Decide, for each week, how many hours you are going to spend on the project, and what you will do in that time. Bear in mind events such as Christmas and exams which will prevent you from spending any time on your project. You will need to spend more time in other weeks to compensate for this.

2.3. Gathering your material

When you have the overview of your report, you simply(!) need to fill in the gaps. For each area of your topic – each theorem, each example, each application, etc. – you need to find one or more sources that cover this. Read what those sources say on the subject, digest it, and write your own account based on what you have learnt.

It is important that your project is based on several sources, since the ability to gather material from different sources is an important element of what is being tested in this project. In particular, your project should not be based solely on the lecture notes for one of your modules. You need to demonstrate that you have sought out different sources and faced down the challenges of working with different sources.

2.4. Appropriate and inappropriate sources

When selecting sources for your project you should consider their level and reliability.

A book may be aimed at undergraduates, or A level, or lower level school children, or at graduate level, or may be a “research monograph” aimed at professional researchers in the field. Clearly books for graduates or researchers will generally be hard to use, and easy to misunderstand. On the other hand, books for A level students or lower may be much easier to understand, but they will usually oversimplify matters. If you present work that is based on an A level book, then it is likely to be of A level standard, and will attract a very low mark when judged at senior undergraduate level.

Textbooks from reputable academic publishers (e.g. Springer, Macmillan, World Scientific, University Presses, AMS, LMS) are highly reliable, as they go through fairly rigorous checking before they are published. Similarly, material on tightly controlled websites (e.g. mathworld.com) can be relied on fairly well.

Most material on the web, on the other hand, is much less reliable. Lecture notes from other universities may not have gone through much checking at all, so may be completely correct or may contain many errors. Wikipedia can be edited by anybody, whether someone knowledgeable on the subject, or completely ignorant. Some entries on Wikipedia are superb, but others are atrocious. Often even the knowledgeable contributors just want to show off what they know so the result is hard to read and often incoherent. If you are not an expert in the subject then it is often hard to tell whether you are looking at a good page or bad. The referencing is a clue: Any reliable facts on a Wikipedia page should be referenced in a proper, reliable source. You should then follow up these references, to find a reliable source instead of citing Wikipedia.

Similar comments apply to youtube videos. There are some superb videos out there containing accurate information presented well. But there are many which are either produced out of ignorance or a wilful desire to mislead. Unfortunately, most do not have any sort of references, so it is much harder to tell good from bad.

If you search for a mathematical topic on the internet you might hit a reliable source but the chances are slim. There are many more unreliable sites out there, and in many cases you will find sites that are just mirrors of Wikipedia, e.g. Answers.com. Even among the reliable sites many are at a completely inappropriate level for an undergraduate project, e.g. BBC bitesize, which is aimed at school children.

2.5. Dealing with material from multiple sources

To take material from different sources, and combine them together into one coherent whole, demonstrates a much greater effort and understanding of the "literature" (i.e. the sum total of knowledge on a subject that has been committed into print). But it is not straightforward to do this; there are often conflicts of notation, and sometimes terminology.

A simple example would be where one book may talk about the function $f(t) = sin(t)$, whereas another may always use x as the independent variable and talk about $f(x) = sin(x)$. If you mix and match then your work may not make sense. For example, if you end up writing "Since we know that $f(t) = sin(t)$, we can differentiate to find that the rate of change is $cos(x)$." A more complicated example might involve both books using the same variable x but to refer to different things.

Terminology in mathematics is *usually* consistent and most words have a fixed definition that all agree on. However, some words are used in different ways in different fields – for example the word "space" might sometimes be used as short- hand for "Hilbert space" in one book, whereas in another "space" means a "topological space", and another might use it as a short form of "vector space".

With definitions there are three particular problems to look out for when you combine material from different sources: 1) you may find the same word used to define different things, 2) you may find multiple different names to define the same thing and 3) you may come across different definitions for the same thing! Let's look at each of these in turn.

1) For example, some authors include zero in the natural numbers while others do not. This gives two definitions of the natural numbers which define different objects. Same name, but different meaning.

2) Some authors refer to s-algebras when discussing measure spaces and some refer to s-fields, but they mean the same thing. Same meaning, but different names.
If you base part of your work on one book and write about s-algebras, and then base another part of your work on a different book and use the term s-fields, then the reader will assume you mean different things, unless you explicitly say that these two terms mean the same thing.

3) A good example of different definitions for the same thing arises with the exponential function. Some books define this as the solution of the differential equation

$$dy/dx = y, y(0)=1.$$

Other books define the exponential function to be the function given by

$$1 + x + x^2/2! + x^3/3! + ...$$

A book that uses the differential equation as the definition will then prove other properties from this equation, such as the Taylor expansion. Such a book might well have a theorem proving that if y satisfies that differential equation, then y has a Taylor expansion of such and such a form. On the other hand, a book that uses the power series as the definition will prove other properties starting from that series. Such a book will probably have a theorem stating that the function given by this series satisfies the above differential equation. If you combine material from the two books without enough care and attention, then you may end up with two conflicting definitions, or proofs that are circular and not logically complete.

A related example of such circular logic comes in the connection between the exponential and the log function. You can either define the exponential directly (e.g. using a differential equation, or power series), and then define the log function to the inverse of this, or you can define the log function directly (e.g. as the integral of $1/x$) and then define the exponential as the inverse of that. What you cannot do is both define log as the inverse of the exponential and define the exponential as the inverse of log; if you do that then neither of them are properly defined. Again, if you combine material from two books that take these two different approaches, then you need to be careful not to end up with such circular definitions.

When this happens, and you mix material from two different sources that use different definitions, the problems can be very difficult to detect unless you understand the subject very well indeed. But your supervisor should be able to help you spot any errors that arise in this way and advise you about correcting them.

For these reasons it is very important that you are clear and consistent in which definitions you use, even if the sources you have read have picked differing definitions. If in doubt, ask your supervisor.

In all cases you must check the meaning of the symbols and words you use and make sure they are used in a consistent way throughout the dissertation.

2.5. Writing up your work

Sooner or later you must turn your ideas and researched material into a written report. You will need to type the material in, so you will need to decide what software to use for this. Most people immediately turn to Word when they want to type something up, but Word is not very good at producing professional quality mathematics. Virtually all professional mathematics is now printed using the software package LaTeX. You have met LaTeX before in the Geometry module MA-131, but there will be some further lectures on the use of LaTeX. We have also put notes about LaTeX on Canvas, so you can look at these as early on in the project as you like.

Some people choose to keep all their notes in pen and paper form until very late on, and then spend a solid chunk of time typing these up to produce the final written report. Other people prefer to type things in as they go along, building up a file steadily, with no frantic rush at the end to type up. It is up to you how you do this – you need to find the pattern that works best for you.

2.6. Ownership of the work

The written report is to be your work, since you have done the job of planning it, structuring it, researching the material in it, and writing it. For this reason it should have your name prominently on the front; you have become an author!

However, the material contained in your report is also the work of others, others who wrote the sources you've used, others who invented/discovered the mathematics that you have written about. And in some cases you will have used sentences word-for- word ("verbatim") from the original sources. It is essential that you make it clear what exactly is your work, and what is somebody else's.

2.7. The *unimportance* of quotations in mathematical writing

A quotation is where we use somebody else's work *in their exact words.* For example,

"Stately, plump Buck Mulligan came from the stairhead" (Joyce, Ulysses, 1922).

We have not re-phrased that at all (not least because it would be hard to re-phrase and capture exactly what Joyce meant), so it is not our work at all. It is a "quotation", or "verbatim quote". For that reason we put quotation marks around it, and note the author, source, and date.

In some other subjects (particularly in the arts or social sciences) it is very common to use quotes, as these are seen as giving authority to an argument. In mathematics we generally do not need to quote other sources directly. In mathematical writing the authority comes from the validity of the logic we apply and the usefulness of the application of our work. In other subjects it is important to use many quotations so as to show how much you have read. In mathematics this is not true; it is better to avoid direct quotations as a quotation shows a *lack* of understanding since you are unable to explain the idea in your own words.

As a rule of thumb, if you quote someone else's work verbatim (i.e., without changing any of the words), then you should assume that this will add *nothing* to the mark you will achieve, except in that it might help the structure and flow of your dissertation. In the light of this, you might reasonably ask whether or not you should include the verbatim quote at all. Often the correct answer is that no, you should not. You should only include such material if it is helpful to the structure of your dissertation.

2.8. Definitions, Theorems, Proofs and Figures

It is a fact of life, however, that definitions and theorems often cannot easily be re- phrased. Definitions, in particular, very often need to be written in a certain way in order to convey the correct meaning. This is accepted, and you do not need to put quotation marks around definitions. Nevertheless, you should give a clear indication of where you have taken the definition from. An example of good practice would be:

> The following definition is taken from [3].
>
> Definition: A *linear transformation* is a function, between two vector spaces, that preserves addition and scalar multiplication.

Alternatively, at the start of a section of your dissertation you could write "All definitions in this section are taken from [3]."

Theorems, Propositions, Lemmas and Corollaries also often need to be worded in a particular way, although there is usually some more flexibility than with definitions. If you use the original wording, you should tell the reader where this has come from. But if you can re-word the statement (correctly!) then this will be counted to your credit, as it demonstrates a better understanding. Of course, if you re-word things incorrectly, then it equally well demonstrates your lack of understanding. This might be something you want to check with your supervisor beforehand, within reason. Even if you do reword a Theorem/Proposition etc then it is important that you reference what the original source was by for instance stating "The following Theorem is based on Theorem 3.1 from [2]" to acknowledge where you are taking your ideas from.

Proofs do not need to be worded in a particular way, so you should try to write these in your own words to demonstrate your understanding of the arguments involved. Even then, however, your proof will almost certainly be *based* on one of your sources, and you should note this. You could write something such as "The following proof is based on the one given in [4]." Often a proof in a book will have many steps which are only sketched or skipped over in order to save space. To demonstrate your understanding you should be looking to fill in these gaps, and show that you know how these missing arguments work.

If you can come up with an alternative proof, using different mathematical ideas, then this would be very strongly to your credit, but most students will not be able to achieve this in many cases, as it is generally very difficult if not impossible!

If you end up copying a proof out word-by-word from one of your sources, then you must indicate this very clearly, by putting the whole proof in quotation marks or clearly marking it as copied. For example you might introduce it with a sentence such as "This proof is copied directly from [4]." Bear in mind that if you have copied a proof word-for-word like this, then you have provided absolutely no evidence that you understand any of the content of the proof. So you have to ask whether it is worth including, since the main purpose of the dissertation is for you to demonstrate what you know and what you understand. If you are thinking of simply copying verbatim an entire proof it would probably be better to leave the proof out and provide the reader with a reference to the proof with a statement such as "A proof of this theorem can be found in [3]." This would save you space to use elsewhere in the project on your own work which will demonstrate your understanding of other aspects of the project and help improve your mark to a greater extent than including copied text.

Some students worry that if they re-write a proof, then it will no longer be correct, and that their mark will be lowered as a result. This generally is not the case. Suppose we have 3 students, A, B and C, doing the same topic, and including the same proof. Student A just copies the proof word-for-word from the book (but referencing it appropriately, of course). Student B re-writes it in their own words, but does not fully understand it, so what they write contains some errors but also shows some understanding. Student C re-writes it, and understands it well, so they are able to write a complete and correct version of the proof. Obviously student C will get the best mark. But of the other two students, student B has demonstrated some level of understanding, but student A has demonstrated no understanding at all. So, based on this proof alone, student B will get a better mark than student A.

Of course, it is not usually so simple. Student A may have copied this proof, but elsewhere in their dissertation, they have given plenty of evidence of a high level of understanding. In that case, they may still get a high mark, even though the copied proof does not add much to their work. Similarly, student B may have shown that they misunderstood this proof, but other things they have written may show that they really do understand much of the subject. And student C may, elsewhere in their dissertation, have included some text copied from a book without any attribution or quotation marks, and so may be guilty of unfair practice and get no credit at all for their excellent re-writing.

Finally, figures and diagrams can be copied from elsewhere, but again you must clearly indicate the source. Alternatively, you may be able to use Mathematica, Matlab, LaTeX, or any other computer package, to produce an equivalent figure of your own. This will be counted to your credit.

2.9. Referencing and plagiarism

The University is very keen to ensure that students are given an appropriate mark for their work. The mark should be *appropriate* for the level of knowledge and understanding that the student demonstrates. And it should be a mark for *their work* - that is to say, a student should not gain credit from other peoples' work.

Students who attempt to pass off other peoples' work as their own are guilty of 'academic misconduct' and liable to a range of punishments depending on the severity of the offence, from a penalty being applied to their mark on the individual piece of work, through to being withdrawn from the University.

You should therefore be very concerned to avoid committing academic misconduct, and this guide is designed to help you avoid it. We give several examples and specific advice below about academic misconduct and referencing in the context of mathematical writing. But there is one overarching principle which will help you avoid unfair practice in your own work:

It should be <u>completely clear</u> which parts of your writing are your own, and which are from other sources.

Someone else should be able to read your work and be in no doubt which parts you have thought up or written yourself, which parts you have taken directly from other sources, and which parts are based on other sources. If your work satisfies that criterion, then you can be reasonably confident that you are not committing unfair practice.

Of course, a safe way to avoid unfair practice would be to take your entire dissertation from another source, and put it all within quotation marks, and say where it has come from. Such an extreme case would avoid unfair practice, but would not deserve a high mark, since the student's *own* work has demonstrated very little knowledge or understanding of the material.

To give a clearer idea of how to use other sources and reference them appropriately, we will go through some more realistic, and less extreme, examples and consider them from the point of view of unfair practice and of what sort of mark they should get.

For these examples we will look at the made up subject of "teaology" which will be the science of how to make a cup of (English) tea. This subject was first discussed by Mrs Bloggs in her seminal book "On the making of a cup of tea" from which we will quote.

Example 1:

> In Bloggs's original work, she described teaology as being "the study of drinking tea as a philosophical position" [2]. She went on to write that "although many aspects of tea-drinking have previously been considered, such as the ultimate digit positioning, the necessity or otherwise of a saucer, and questions of pre-lactation or post-lactation. Hitherto, however, the philosophical implications of drinking tea have been insufficiently addressed".

It is clear from this what text is taken from the source, namely the words between the quotation marks, and what is not. It is perfectly clear what the source is, namely item 2 in the bibliography. And it is perfectly clear who wrote the words being quoted.

This is an example of good referencing practice. Nevertheless, there is little content in the paragraph that is the student's own work, so this paragraph gives very little evidence of understanding or knowledge. Such evidence is needed in order to justify a good mark, so this dissertation would get a low mark unless evidence of understanding and knowledge is given elsewhere in the dissertation.

Example 2:

> Bloggs described teaology as being the study of drinking tea as a philosophical position and that many aspects of tea-drinking have previously been considered, such as the ultimate digit positioning, the necessity or otherwise of a saucer, and questions of pre-lactation or post-lactation. Hitherto, however, the philosophical implications of drinking tea have been insufficiently addressed.

Now it is clear that some reference is being made to Bloggs's work, but it is no longer obvious how many words have been written by the author, and how many by Bloggs. This is not unfair practice, since the student is clearly attributing *some* of this to Bloggs, but it is an example of poor referencing, since it is not very clear *what* is being attributed. The implication of this referencing is that the ideas have come from Blogg's but the text is the student's own interpretation. However, we can see from the quotations in Example 1 that this is not the case. Turnitin will identify when text has been copied like this and bring it to the markers attention. This work also shows poor grammar, as a result of being copied-and-pasted from the source in a careless way.

Example 3:

> Teaology is the study of drinking tea as a philosophical position and many aspects of tea-drinking have previously been considered, such as the ultimate digit positioning, the necessity or otherwise of a saucer, and questions of pre-lactation or post-lactation. Hitherto, however, the philosophical implications of drinking tea have been insufficiently addressed.

In this example no reference is made to Bloggs's work, and there is no indication that any of these ideas or any of these words were written by someone other than the student, nor is there any demarcation of which words were written by the student and which by someone else. The reader would naturally assume that these words are written by the student themselves. So this constitutes unfair practice because, in effect, the student is passing off others' work (namely Bloggs's) as their own. Since turnitin very clearly highlights such things, it is easy for us to spot things like this. Effectively here the student is attempting to deceive the marker. The penalty that would be applied in this case would depend on many factors but it could include a zero mark for the whole dissertation or even for the whole year of study.

Example 4:

> The work in this chapter is based on Bloggs [2].
> ...
> Teaology is the study of drinking tea as a philosophical position and many aspects of tea-drinking have previously been considered, such as the ultimate digit positioning, the necessity or otherwise of a saucer, and questions of pre- lactation or post-lactation. Hitherto, however, the philosophical implications of drinking tea have been insufficiently addressed.

This is very poor. There is a reference to Bloggs's work which acknowledges where the ideas have been taken from, but this is a very indirect reference. This would be appropriate if the rest of the section contained re-phrased versions of what Bloggs wrote, but the text here is mostly copied word for word, with no indication that some of these words were written by someone other than the student. This constitutes poor referencing, and suggests unfair practice. It will be detected by turnitin and highlighted as being taken from another source to draw this matter to the attention of the marker.

Example 5:

> Bloggs described teaology as the subject of drinking tea by way of a philosophical position. Some things about tea-drinking have previously been considered, for example, ultimate digit positioning, the need or not of a saucer, and problems of pre-lactation or post-lactation. Later, but, the philosophical implications of drinking tea have been not well enough considered.

Now we see that the student has made some effort to re-write Bloggs's work in their own words. Since Bloggs's work itself is referenced, it is not unfair practice. But the re-wording is very clumsy, re-using many of the original phrases in a way that suggests the student does not understand their meaning. So although this is not unfair practice or especially poor referencing, it is simply not very good, and would not justify a good mark.

Example 6:

> Teaology is the subject where one considers drinking tea in a philosophical way. Several different aspects of drinking tea have been studied before, including how you should hold your little finger while drinking, and whether you need to use a saucer or not, and whether you put milk in first or after the tea. But nobody before Bloggs had looked at the philosophy of drinking tea.

Now there is little reference to Bloggs's work, but the original words have been read, understood, and re-phrased in the student's own words. This demonstrates a very good level of understanding and provides evidence that would justify a good mark. No text has been quoted without attribution, so in this respect there is no element of unfair practice. However the student should have acknowledged that the ideas being discussed here were taken from Bloggs's book by including some reference to this source.

Example 7:

> To paraphrase Bloggs [2], teaology is the subject where one considers drinking tea in a philosophical way. Several different aspects of drinking tea have been studied before, including how you should hold your little finger while drinking, and whether you need to use a saucer or not, and whether you put milk in first or after the tea. But nobody before Bloggs had looked at the philosophy of drinking tea.

Here Bloggs's work is clearly referenced, but the student has re-phrased that work in their own words. This demonstrates a very good level of understanding, and excellent referencing.

Example 8:

> In [2] Bloggs wrote that "teaology is the study of drinking tea as a philosophical position"; in other words, teaology is the subject where one considers drinking tea in a philosophical way. Several different aspects of drinking tea have been studied before, including how you should hold your little finger while drinking, and whether you need to use a saucer or not, and whether you put milk in first or after the tea. But nobody before Bloggs had looked at the philosophy of drinking tea.

This is again very good, in that Bloggs's own words are recorded, and clearly marked as such by the use of quotation marks. But also the student has re-phrased that work in their own words. This demonstrates a very good level of understanding, and perfect referencing.

2.10. Your bibliography

At the end of your written report you should list the sources you have used. This list is the "bibliography" for your report. The sources should be listed in alphabetical order by the author's surname, and each source should be given a label (typically a number in square brackets), so that in your report you can refer to that source simply by that label.

For example, you may write that the following theorem is taken from [4]. Alternatively, you may wish to be more specific, since [4] is a large book, so you may want to write "in [4], p. 234, we find ...." , or "in [4], Proposition 3.25, we have ...".

In each case the principle is to allow your reader to find your sources, so that they can read the material in its original context. This ability to trace material back to source is a key part of the credibility of academic writing. If you give no sources, then many people will simply not believe what you say.

For each source you need to give enough information for your reader to find the source. The necessary information is different according to whether the source is a book, a website, a set of lecture notes, or something more transient. Below is a typical bibliography (although it is too short for a typical report) with examples of different types of source – a journal article [1], a set of lecture notes [2], a website [3], two printed books [4] and [6], and one online book [5]. Note the different information provided for each – publisher for the books, visit dates for the websites, etc. Note also that most of the sources listed there are printed sources, not websites. This means they have been through a robust checking process and can be trusted. It is acceptable to use some websites as sources, but they are not as reliable, so you should have more printed sources than you have websites in your bibliography.

References

[1] Bochner, S., Diffusion equation and stochastic processes. Proc. Natl. Acad. Sci. U.S.A. 35 (1949), 368-370.

[2] Crossley, M.D., Applied Algebra: Coding Theory. Lecture notes, Swansea University, 2015.

[3] Fleming, G., Unreliable Sources for Your Research Project. https://www.thoughtco.com/bad-research-sources-1857257 Visited September 23, 2022.

[4] Hatcher, A., Algebraic Topology. Cambridge University Press, 2001.

[5] Houston, K., How to Write Mathematics. http://www.kevinhouston.net/pdf/htwm.pdf Visited May 26 , 2017.

[6] Knuth, D.E., Surreal numbers. Addison-Wesley, 1974.

2.11. Length

The written report (excluding bibliography and any appendices containing Matlab code etc.) is required to be between 7000 and 8000 words in length. This may correspond to 20 pages of fairly dense text, or up to 50 pages if there are more figures, pictures, displayed equations, or lots of empty space on the pages. The length, in terms of the number of pages, does not matter so long as the word count is in the required range. We recommend that you use TeXcount, which is tool that is specifically designed to count words within LaTeX documents. The web-based version of this tool can be found at

http://app.uio.no/ifi/texcount/online.php

Chapter 3: Secondary Goals

3.1. Introduction

The written report is the primary goal of this project, but it is not the only goal. There are a number of secondary goals.

Some of these goals are to help you plan and structure your final report, whilst others are to help you develop other skills such as presenting your work.

The six secondary goals are:

1. To complete some initial reading provided by your supervisor and then present and discuss your work and plans with your supervisor.
2. To write a "Project Plan" summarizing your plan to manage the project.
3. To submit an "initial draft" of a section of your report so that you may receive some useful feedback on your writing.
4. To give an "interim presentation" on your achievements part way through the project.
5. To reflect upon what you have learned, how you worked, and how you adapted your original "Project Plan".
6. To give a "final presentation" on your final written report once this is completed.

Each of these secondary goals has its own deadline which you can find on the project calendar.

All of these secondary goals are important, demonstrating significant skills which are an essential part of the module. For this reason

Failure to complete any of the secondary goals will lead to a mark of zero for the module.

3.2. Secondary Goal 1: The initial reading and discussion

The first goal for the project is to complete the initial reading and tasks that your supervisor will provide you with in your first meeting. The aim of this initial reading and tasks are to introduce you to the topic that you will be working on. Before you can plan what you are going to cover in your project you need to get an overview of the area you are working in and begin to get to grips with the terminology used in that subject. The only way to do this is to start reading for yourself!

You will need to complete the reading and tasks that your supervisor outlines for you before the first deadline, at which time you will need to give a short presentation on what you have learned to your supervisor. Your supervisor will then discuss this with you and help you to draw up some outlines for your project plan. Our aim here is to ensure that you have started working on your project and have a clear idea of where you are going with your project.

3.3. Secondary Goal 2: The Project Plan Form

Once you have completed your initial reading and discussed your ideas with your supervisor, you will need to draw up a plan of action for the rest of the year.

It is your project so you will need to decide what the content is going to be, how much space you are going to allocate to each topic, what the overall aim and end point of your project is going to be, what you are hoping to cover etc. Your supervisor will guide and advise you in terms of what is sensible and achievable, but they will not tell you exactly what to include. It is your project and you need to take responsibility for it!

The project plan will consist of your plans in terms of content and structure for the project and it also needs to outline the timeline for the project. A good plan should have clear measurable goals at regular intervals so that you can easily assess if you are on target for completion.

3.4. Secondary Goal 2: The initial draft

You will be required to submit an initial 4-page section of written work at the end of semester 1. This is an opportunity for us to give you feedback on what you have written, to note any errors and points for improvement.

This feedback will identify exactly 4 points of your report that could be improved in order to get a better mark. The aim is to help you improve the project, not to identify every single deficiency in your work.

Please bear in mind what the feedback will *not* do:

- The feedback will not tell you what you should write instead but will guide you as to what is wrong or needs to be improved.
- The feedback will not tell you exactly what you need to do to achieve a perfect mark, since there is no such thing as a perfect project.
- The feedback will not tell you exactly what you need to do to achieve a “first class” mark for your project.

Once you have received your feedback, you should think carefully about it and then use it to improve your draft.

It is important to ensure that you fully understand what the feedback is highlighting. It will not necessarily be easy to incorporate your feedback into your project, and may take quite some effort as well as understanding. You can only improve your report if you understand what is going on. It may be that through misunderstanding the topic you might not improve your report very much and, in some cases, you may even make your report worse. It is also important to consider if a point raised has implications elsewhere in your report; you may have repeated a mistake throughout the project.

Ultimately the mark you get for your project will not be simply a reflection of the effort you put in – it will also be strongly influenced by the knowledge and understanding you demonstrate in what you write.

3.5. Secondary Goal 5: The final reflection

You will be required to write a 500-700 word section at the end of your project report about your reflection on the process of completing your project. This section will be entitled “Final Reflections”, and will be placed just before the References in your report.

Your report will be largely assessed on the mathematical accuracy of what you wrote. This is reflected in the way you structure your report and in the way you present and explain the mathematical facts which are relevant for your project. But you will end up with the final version of your project report as a result of a process of independent work which will not always have been straightforward. The challenge in finding extra sources of information and references on aspects of the topic you need to understand better, is not an easy task. Understanding what you read from more advanced sources, and more specialized books is often far from easy. For this it may be necessary for you to revisit multiple times what you have already read, and to have further thoughts on it, possibly trying to find your own examples and trying your own calculations. All these attempts, especially when they are on a particularly challenging topic, may involve mistakes that you will spot by revisiting them and re-thinking about them. Do not be afraid of making mistakes while you are learning, but rather, you should regard them as an almost essential step to improving the quality of your understanding. As you develop familiarity with the topic you should be able to go back and spot mistakes and correct them. This is the normal way to learn mathematics at all levels, including when doing research in any mathematical field! For this reason you should attempt work that challenges you as evidence of your independence and personal initiative, particularly in tackling issues which are not entirely covered in the reading directly suggested by your supervisor.

Your Final Reflection should help to emphasize the importance of a continual process of self-reflection on the boundaries of your knowledge, to consolidate what you know already and fill as many gaps as possible in it. They will detail the way you have learnt from your own struggles (e.g. referring to one of your own specific mathematical mistakes: What were you trying to do? What for? Where was the mistake? What did you learn from it?) In this section you can reflect on the Project Plan you have submitted at the beginning of the year, writing about the actual skills you have developed and if and why you had to recalibrate the goals you originally have identified in the Project Plan. We will assess this particular section on the way you critically reflect on the learning path while working on the project, rather than on whether you have achieved all the targets you set at the beginning of this mathematical journey.

3.4. Secondary Goals 4 & 6: Presentations

You will give two presentations during the course of the project – an interim presentation mid-way through the year, and a final presentation after you have completed your written report. At the start of the year we will give a clearer idea of which week these presentations will take place, and we will publish the exact details of when, and where, a week or so before they are scheduled to take place.

In both presentations your aim is to show some of the mathematics that you have learnt so far. The focus is not about you and your achievements, but rather the focus is on the mathematics itself. Of course, what mathematics you present in the interim presentation will give us an indication of your achievements so far, but that is not something to worry too much about – in the presentations we are not assessing how much maths you've learnt, but rather how well you can present mathematical material.

Standing up in front of a group of people and speaking about your project can be a daunting task. Most people would feel nervous in such a situation. However, it is a very useful skill to develop. Most of you will already have given some presentations, for example in the Geometry module MA-131, and you can look back on these and reflect on how well you did and how you can improve. By the end of the project, you can confidently look back knowing that you have successfully delivered two more presentations, and you will probably find they were easier because of your increased experience. So when you are next asked to give a presentation on your work, you can have still more confidence and feel less nervous, because of your further experience of doing this.

For any presentation there are several things to bear in mind: the audience, the time allotted, the purpose of the presentation, among others. You should plan your presentation carefully so that you can convey the ideas and information within the time available, and in such a way that your audience will follow them. We will give more guidance about this in lectures before each of the presentations.

As well as planning, you should practise. Gather a few friends together in the week leading up to the presentation and run through your presentation with them. Whether they are mathematicians or not they will be able to spot strengths and weaknesses in your presentation, and give you constructive feedback. And if they have no feedback, then you will still have benefitted from the practice – once you have done the presentation once you will be more comfortable the next time. You may even wish to practise your presentation several times, so that when you do it for real, you are as relaxed as possible.

At the end of your presentation there will be an opportunity for the audience to ask questions, so you should be prepared for this. Obviously you cannot prepare the answers since you do not know the questions in advance. But the questions will normally be about the material you have presented, so you should be able to answer them based on what you have learnt. And if you cannot, then just answer that you do not know. Your response to the questions will not affect your work for the presentations, although if you are asked several questions and your answer to every one of them is “I don’t know”, then this may reflect badly on you.

3.5. Differences in the two presentations

The two presentations are structured differently, to reflect their position in the overall project timeframe, and your developing confidence at giving presentations.

The interim presentation is shorter (10 minutes), and delivered to a smaller group of people – typically 5 or 6 students and 1 or 2 members of staff. This presentation is usually given in a lecturer’s office, using a whiteboard. If it is not possible to do this in person it may instead be done online.

The final presentation is longer (15 minutes), delivered to a larger group of 20-25 students and 2 members of staff. This will be given in a lecture room and should be delivered using a data projector. If it is not possible to do this in person it may instead be completed online.

You will develop the ability to speak in front of a smaller group first, and then a larger group. You will develop the ability to present using first a whiteboard, and then later, using powerpoint or an equivalent better suited to mathematics.

In each case you should think in advance about what you will write up on the whiteboard, or put on the slides, and what you will say. These should not be the same – you should not just read from the whiteboard or screen – but they should complement each other.

With the whiteboard you have the flexibility that you can change what you write, even as you write it. But you have the challenge that when you write you are naturally facing the board, whereas if you are speaking, you need to face the audience and speak into the room, with a good, clear voice.

With a computer-based presentation, you have the advantage that you prepare the written material in advance and so you do not need to worry about that during the presentation. Of course, that means if there are any mistakes, then you are stuck with them during the presentation, so it is best to check your slides as carefully as possible before the presentation.